Fernando Román Hidalgo

AI & Robotics Engineer
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Summary

AI and robotics engineer with a **multidisciplinary profile and systemic vision**, featuring professional experience in developing AI agents, MCP integrations and workflow automation, built on a solid foundation in robotics, control and embedded systems. I possess a **high capacity for abstraction in solving complex problems** and **accelerated self-taught learning**, enabling me to provide significant value in technically demanding projects. My objective is to consolidate an ambitious career plan at the intersection of **artificial intelligence and physical systems**, within dynamic and innovative environments where continuous improvement is the standard.

Experience

AI Engineer

Atlantica Sustainable Infrastructure | Seville, Spain

Generative AI, Agents, MCPs and Workflow Automation — August 2026 - Present

- **AI Workflow Automation:**
  - Design and implementation of automated workflows with n8n and language models, replacing manual processes based on Excel, email and scattered documents.
  - Automatic ingestion and processing of emails and documents (PDF, SharePoint) with structured information extraction using AI.
  - Integration of workflows with the corporate ecosystem: Microsoft 365, SharePoint, Azure DevOps and monday.com.
- **Agents and MCPs:**
  - Development and integration of MCP (Model Context Protocol) servers to connect AI models with corporate tools and data sources.
  - Design and deployment of AI agents with Claude (Claude Managed Agents) for natural language database queries and orchestration of multiple specialized agents.
  - Prompt optimization, per-call cost control and model evaluation to balance cost and quality.
- **AI-Assisted Development (ADLC):**
  - Software development with Claude Code following an agent development lifecycle (ADLC): specification, implementation, review and integration of each requirement.
  - Creation of Claude Code skills and subagents to automate team tasks, such as sprint planning and tracking in Azure DevOps.
  - Fullstack development of internal AI applications (FastAPI, PostgreSQL, React) and deployment to Azure Container Apps with CI/CD.

Software Engineer

Airtificial Aerospace & Defense | Seville, Spain

Fullstack Development, DevOps, Automation and Project Management — April 2025 - May 2026

- **Management and Client Relations:**
  - Project management for the development of a process automation application for a major aerospace company.
  - Direct customer interaction for requirements definition and specification.
  - Technical leadership and coordination in the development of custom solutions.
  - Application of the Agile Scrum framework integrated with Lean principles to optimize workflow and eliminate inefficiencies in development.
- **Technical Development and Automation:**
  - Use of Git for version control in both projects.
- **Web Application:**
  - Development of a web application with Google App Script (JavaScript, HTML, CSS) to replace obsolete Excel spreadsheets with an intelligent and automated interface.
- **Digital Twin and Optimization Platform:**

- Design and implementation of a digital twin for a composite material production plant, with a FastAPI and PostgreSQL backend, React frontend, deployed using Docker containers.
- Production deployment: environment configuration, port opening, Nginx reverse proxy for traffic management, and full system commissioning.
- Development of predictive AI models in Python for early failure detection using industrial sensor data.
- Implementation of complex production scheduling algorithms in Python using Google OR-Tools' CP-SAT solver, with variable capacity-load constraints and custom priority weighting systems.
- Orchestration of data pipelines and execution of AI models with Apache Airflow.
- Integration with SAP for synchronizing master data and production planning transactions.
- Acquisition and interpretation of data from industrial sensors to feed the digital twin and predictive models.

Projects

To see all my projects in detail, check out my **Interactive Portfolio**.

Honors and Awards

Winner — National Competition Robotics League (LNRC) | Spain
Minisumo Category — 2024

Education

Master's Degree in Electronic Engineering, Robotics and Automation — University of Seville, Seville, Spain — 2025 - 2026

Bachelor's Degree in Electronic Engineering, Robotics and Mechatronics — University of Seville, Seville, Spain — 2021 - 2025
GPA: 7.5/10

Technological Baccalaureate (Graduated with Honors distinction) — IES Heliche, Seville, Spain — 2019 - 2021

Certifications

Google Developer Program — Machine Learning — Google Developers — 2026
Completed Machine Learning courses within the Google Developer Program.

AI with Google Gemini — Banco Santander & Google — 2026
Training program on developing and modeling solutions using Google Gemini generative AI.

GitHub Copilot — Software Development — GitHub — 2026
Practical course on AI-assisted software development and workflow optimization.

Advanced Python Course — Idexa — Jan 2026
Online

Certificate in Advanced English (C1 Advanced) — Cambridge Assessment English — Sep 2024

Basic Life Support and Automated External Defibrillation — Ympetu — 2020

Aquatic Lifeguard — Ympetu — 2020

Skills

Programming	C, C++, Python (OpenCV), MATLAB, Simulink, ROS/ROS2, JavaScript, React, VBA, HTML, CSS, SQL, SAP/HANA
Electronics	Digital/Analog electronics, microcontrollers (MSP430, ESP32, Arduino...), RTOS, FPGA, automation and control
Simulation	System modeling and simulation (Gazebo, Webots), PID control design, embedded systems integration
DevOps	Docker, Kubernetes, AWS, Linux, Git, CI/CD, Apache Airflow, Azure (Container Apps, DevOps, Bicep)
AI / LLMs	Language Model APIs (Gemini, Claude, Copilot...), Prompt Engineering, Agents (Claude Code, Claude Managed Agents, skills and subagents), MCP, n8n, Machine Learning applied (LSTM, predictive models), Reinforcement Learning (DRL)
Protocols	TCP/IP, MQTT, Zigbee, CAN bus, UART, I2C, SPI

3D Design	Fusion 360, FreeCAD, KiCad

Soft Skills

Capabilities	Communication, leadership, adaptation, proactivity, critical thinking, punctuality, autonomy, management and teamwork

Languages

Spanish	Native
English	Professional competence (C1)
French	Basic competence